

A Data Visualisation Story on Global Poverty

By Anne Gnanarajan



*Created for Australian NGOs and non-profit organisations
with a focus on global poverty alleviation*

Aim and Supporting Datasets

Aim: To provide a data visualisation portfolio (and story) for Australian NGOs and non-profit organisations to:

1. Help make strategic decisions about where to allocate resources / funding
2. Help educate potential sponsors about strategic priorities

Datasets Used

1. **“Poverty and Inequality Platform”** (World Bank, 2024) – data about poverty rates (by country/year) up to 2022
2. **“World Economic Outlook”** (IMF, 2024) – data about GDP and economy (by country/year) up to 2022

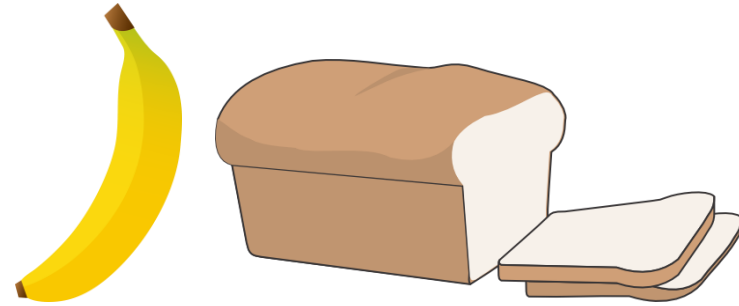
Data was manipulated into a wide table using “Knime Analytics Platform” to feed into “Tableau”.

Addressing Data Limitations

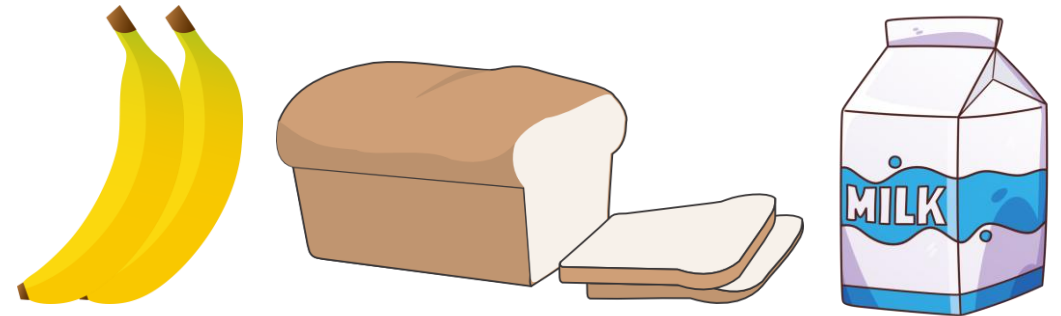
- **Missing values:** *Average of the last 5 years* has been used for “recent snapshot” visualisations to address adhoc missing values for certain years for certain countries.

Poverty Thresholds

\$2.15 per day: The **International Poverty Line (IPL)**, currently set at \$2.15 per day, is a global monetary threshold established by the World Bank to reflect “**absolute poverty**”. (Kenton, 2023)

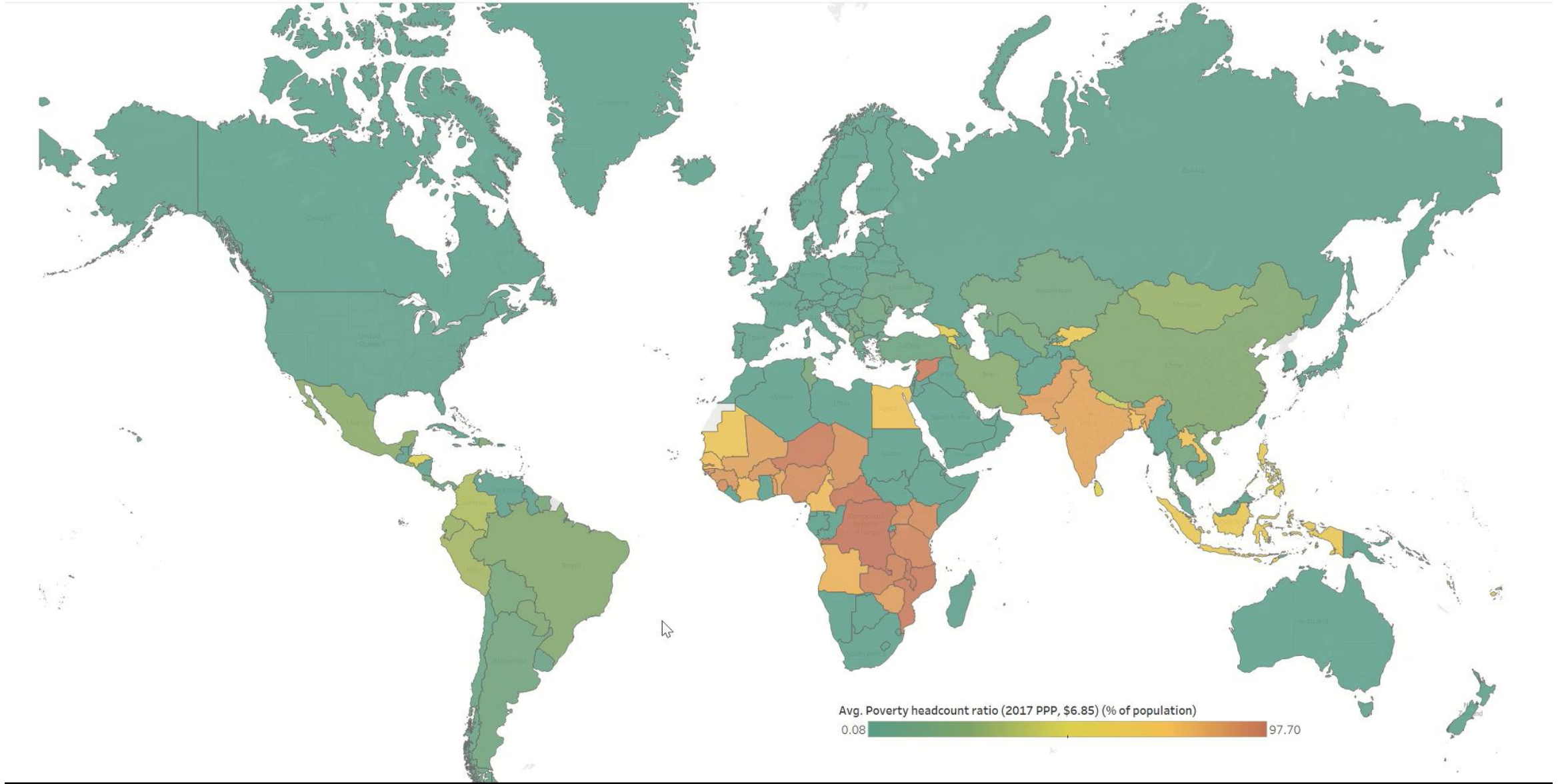


\$6.85 per day: The \$6.85 per day poverty threshold is another global monetary standard set by the World Bank, to reflect the **poverty standards of upper-middle-income countries**. (World Bank, 2025)



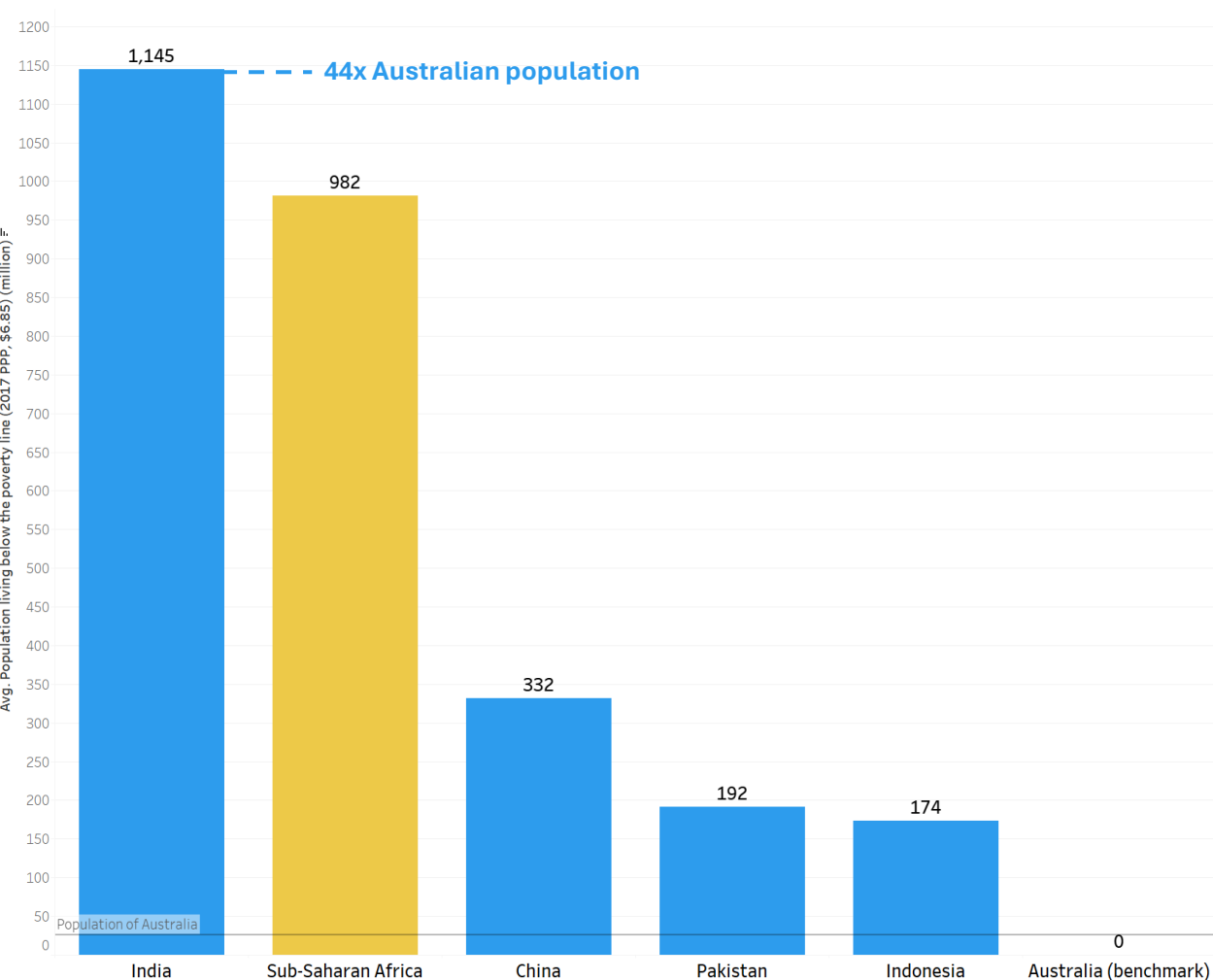
The Global Poverty Landscape: India, Pakistan, Syria and many countries across Africa have over 70% of their population living on under \$6.85 per day.

Percentage of Population living below \$6.85 poverty threshold

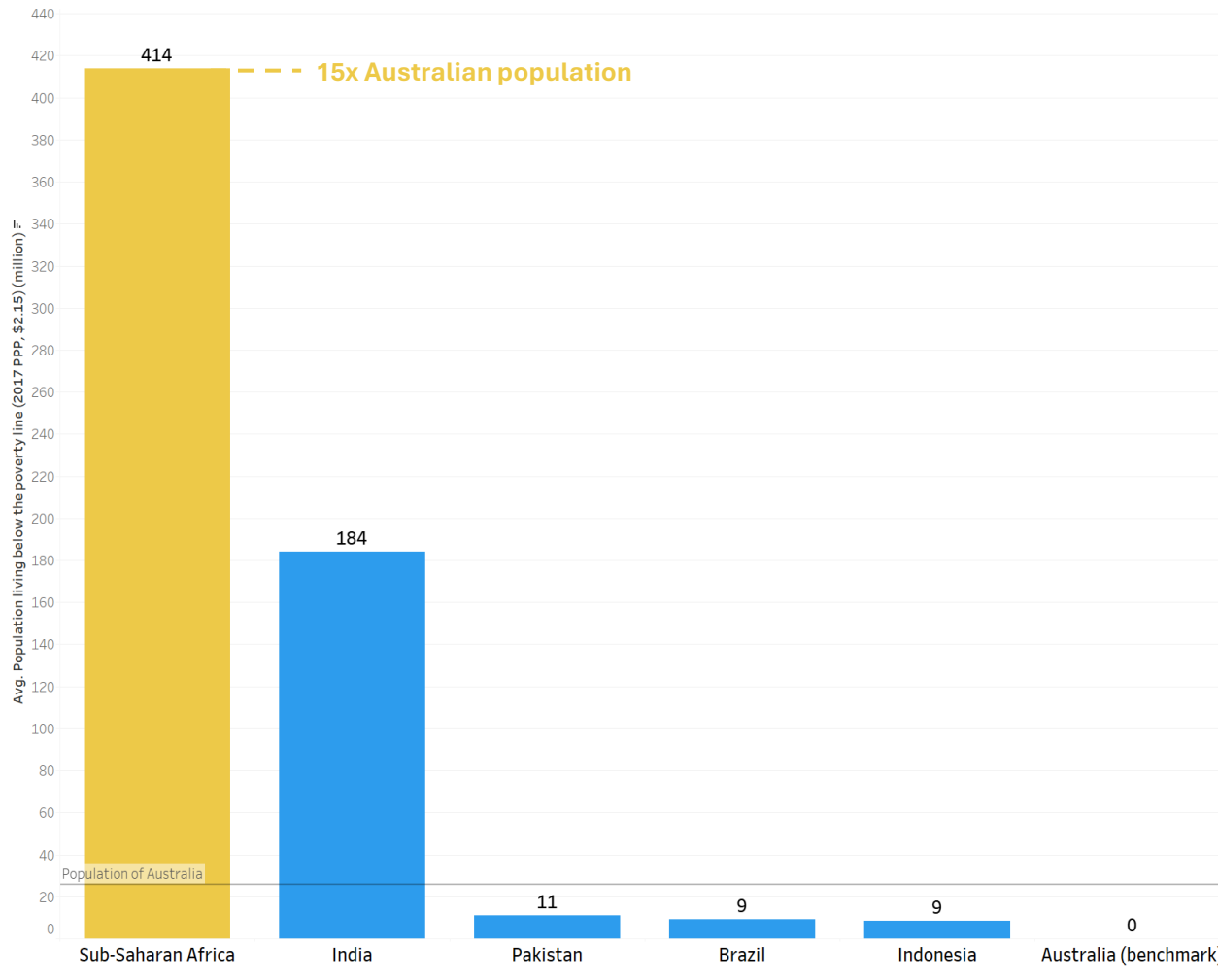


Countries most affected by poverty: Whilst India has the greatest number of people living under the **\$6.85 threshold**, Sub-Saharan Africa has the great number of people living in absolute poverty - **\$2.15 per day**.

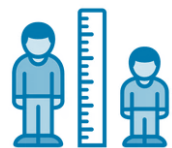
Population of people living below \$6.85 a day – with Sub-Saharan Africa grouped



Population of people living below \$2.15 a day – with Sub-Saharan Africa grouped



Impacts of poverty: There are several impacts of poverty, including undernourishment, growth-stunting, higher infant mortality rates, lower literacy rates, underemployment and increased crime.



31.5%

of people moderately or severely stunted
(Riwa, 2025)

36%

children under 5 years growth-stunted
(Government of India, 2022)

2.3%

children under 5 years growth-stunted
(Global Nutrition Report, 2025)



47 per 1000

infant mortality rate
(WEO, 2022)

25 per 1000

infant mortality rate
(Macrotrends, 2025)

2.3 per 1000

infant mortality rate
(Queensland Family & Child Commission, 2024)



9.4%

Gross Tertiary Education Enrolment ratio
(WorldBank, 2021)

28.4%

Gross Enrolment Ratio for higher education
(Ministry of Education, 2025)

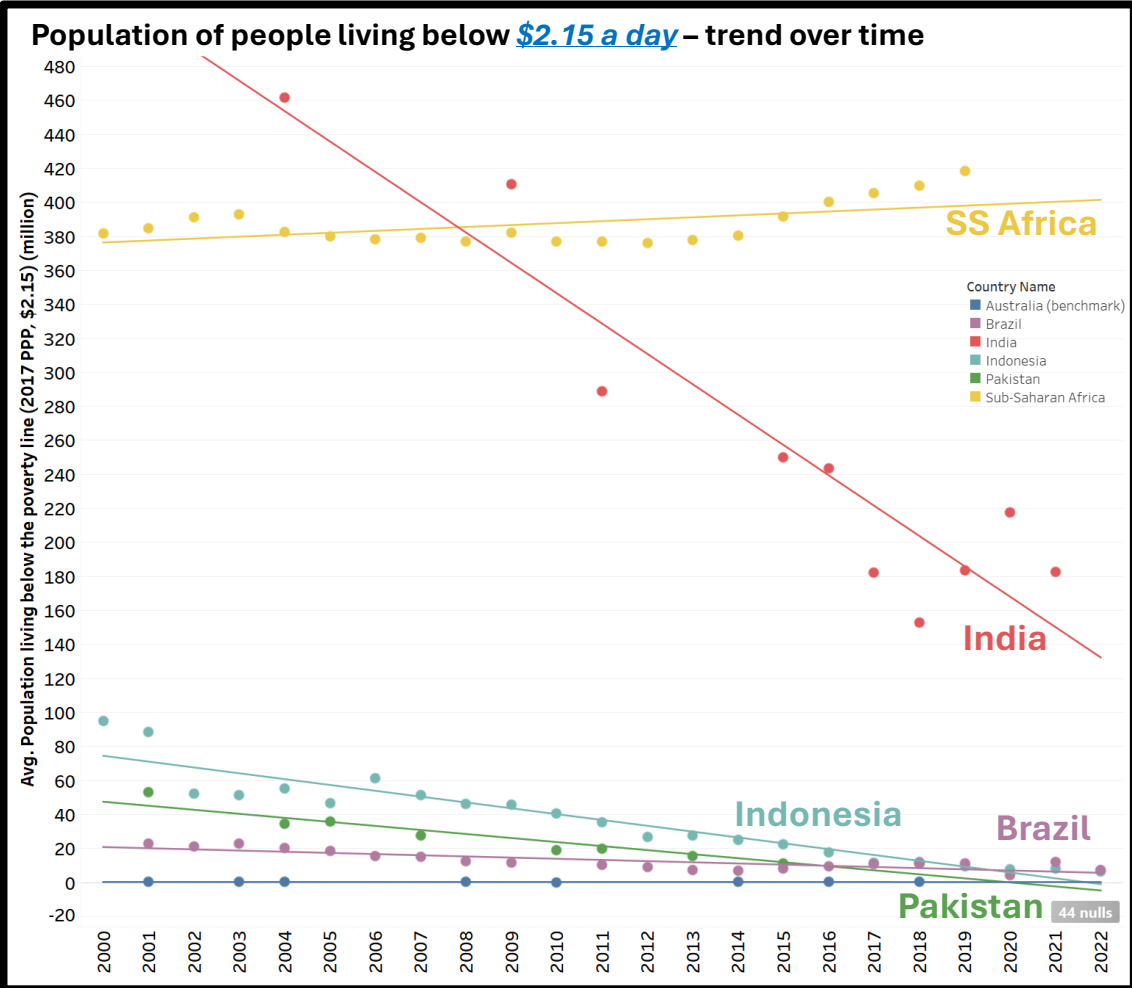
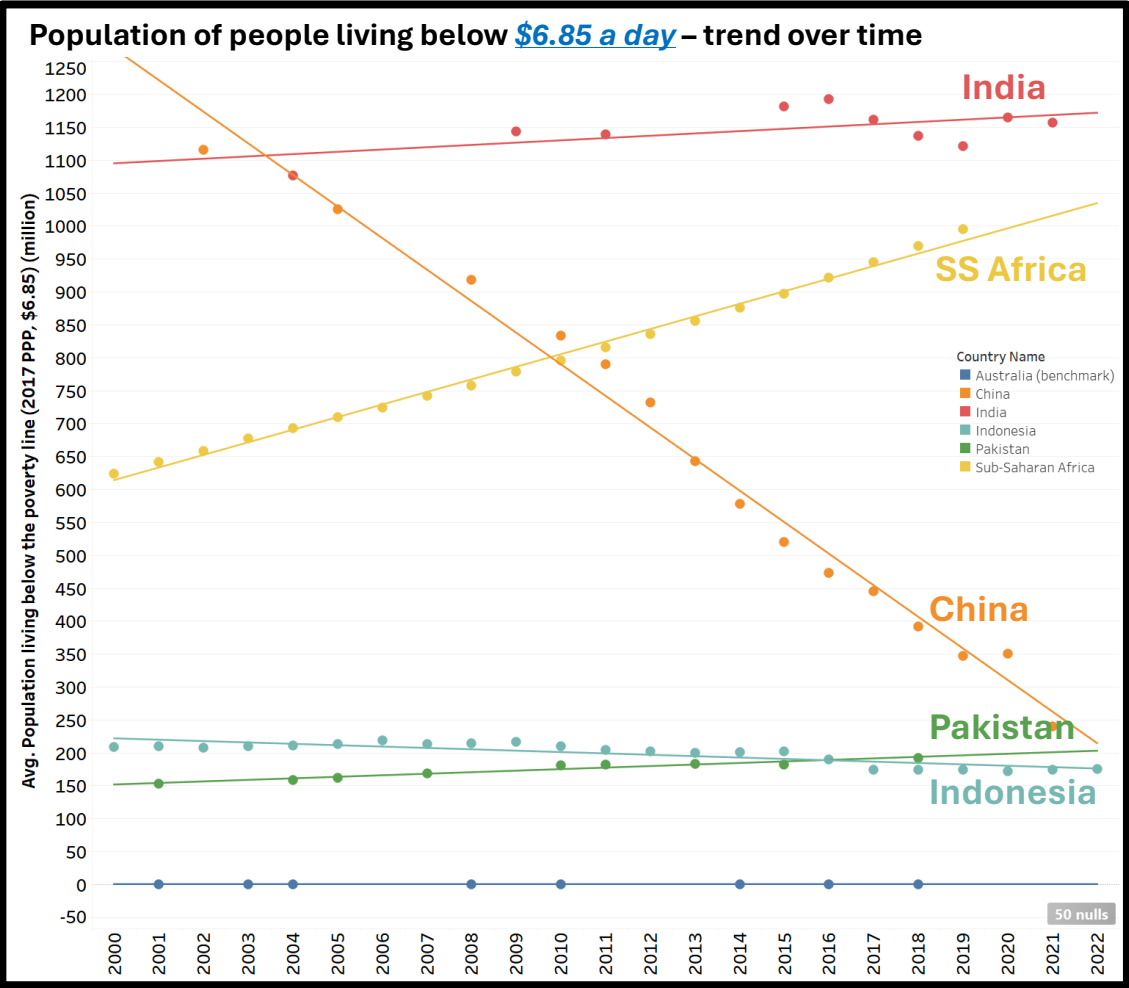
47.2%

Tertiary education attainment for population 25+
(ABS, 2024)

**Is there any
hope for these
countries?**

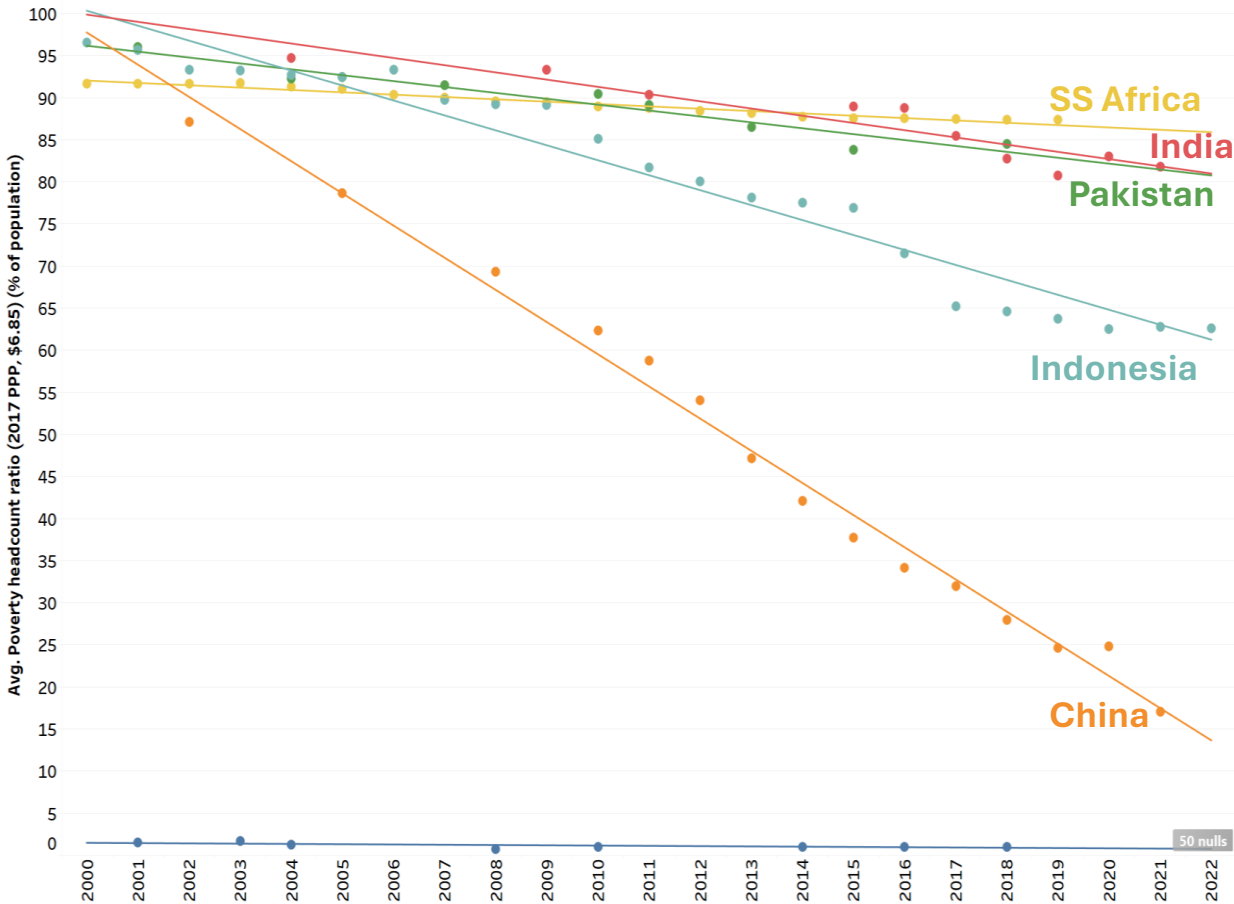


Changes in poverty over time: China has made huge strides in reducing numbers of people living under \$6.85 a day, and similarly, India has drastically reduced its number of people living under \$2.15 a day.

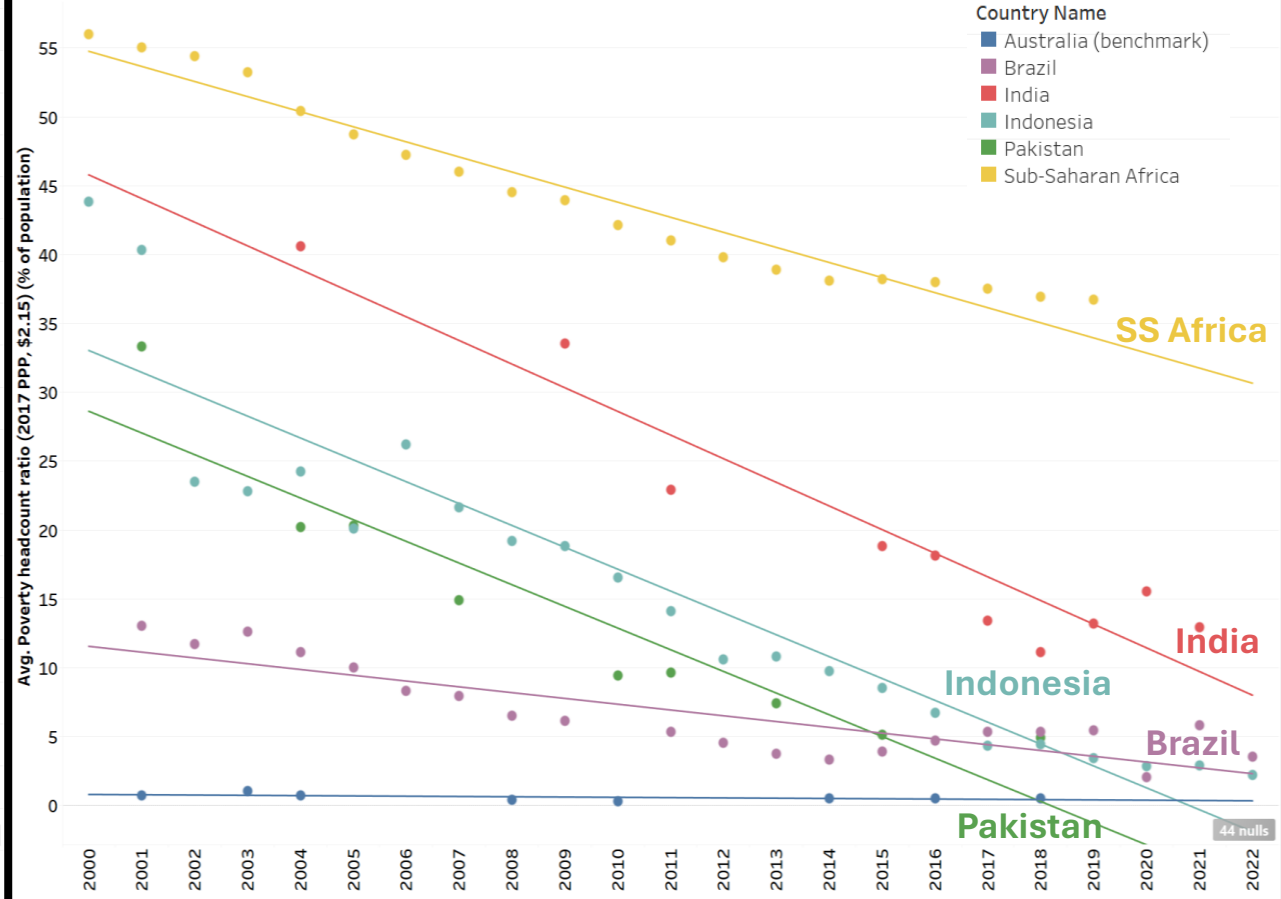


Changes in poverty over time: Looking at both thresholds, all regions, even Sub-Saharan Africa have seen a consistent reduction in the percentage of people living in poverty.

Percentage of people living below [\\$6.85 a day](#) – trend over time



Percentage of people living below [\\$2.15 a day](#) – trend over time



Recommendations for Australian NGOs and non-profits

What	Why	Action
Use Data-Driven Targeting for Resource Allocation and Continuous Monitoring	To ascertain where poverty is most severe and where impacts (like stunting and low education rates) are concentrated.	Incorporate Tableau dashboards into strategic planning sessions to: • Identify high-priority countries/regions • Match intervention types to data (e.g., nutritional programs in areas of stunting)
Engage Donors Through Visual Storytelling	Sponsors respond emotionally to visual clarity and evidence	Use charts in campaigns and reports to: • Explain need and urgency • Demonstrate data-backed impact area
Invest in Integrated Interventions	Poverty's impacts are multidimensional — nutrition affects education, which affects crime.	Launch programs that combine education + health + safety — e.g., school feeding programs that encourage attendance and reduce malnutrition.
Help stimulate local businesses activity	Increased business activity can improve employment prospects for locals, and help build self-sustaining economies	Partner with a list of priority local SME businesses to help stimulate local business activities

References

- ABS, 2024. *Education and Work, Australia*. [Online]
Available at: <https://www.abs.gov.au/statistics/people/education/education-and-work-australia/latest-release>
- Fernando, J., 2025. *Gross Domestic Product (GDP) Formula and How to Use It*. [Online]
Available at: <https://www.investopedia.com/terms/g/gdp.asp>
[Accessed 2 June 2025].
- Figma, 2024. *What are the Gestalt Principles?*. [Online]
Available at: <https://www.figma.com/resource-library/gestalt-principles/>
- Global Nutrition Report, 2025. *Australia and New Zealand*. [Online]
Available at: <https://globalnutritionreport.org/resources/nutrition-profiles/oceania/australia-and-new-zealand/>
- Government of India, 2022. *National Family Health Survey*. [Online]
Available at: <https://dhsprogram.com/pubs/pdf/FR375/FR375.pdf>
- IMF, 2024. *World Economic Outlook (WEO)*. [Online]
Available at: https://data360.worldbank.org/en/dataset/IMF_WEO
[Accessed 5 June 2025].
- Kenton, W., 2023. *International Poverty Line: Definition, Criticism, Uses*. [Online]
Available at: <https://www.investopedia.com/terms/i/international-poverty-line.asp>
[Accessed 18 May 2025].
- Macrotrends, 2025. *India Infant Mortality Rate (1950-2025)*. [Online]
Available at: <https://www.macrotrends.net/global-metrics/countries/ind/india/infant-mortality-rate>
- Ministry of Education, 2025. *India's Higher Education from Tradition to Transformation*. [Online]
Available at: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2102789>
- Queensland Family & Child Commission, 2024. *Child death reports and data*. [Online]
Available at: <https://www.qfcc.qld.gov.au/sector/child-death/child-death-reports-and-data>
- Riwa, F. P. e. a., 2025. *The Prevalence and Determinants of Undernutrition Among Infants and Children Aged 6 Months to 5 Years in Sub-Saharan African Countries: A Systematic Scoping Review*. [Online]
Available at: <https://academic.oup.com/nutritionreviews/article/83/7/e1896/7943446>
[Accessed 15 June 2025].
- UN, 2025. *Ending Poverty*. [Online]
Available at: <https://www.un.org/en/global-issues/ending-poverty>
[Accessed 18 May 2025].
- WEO, 2022. *This is how Africa is reducing its child mortality rate*. [Online]
Available at: <https://www.weforum.org/stories/2022/04/infant-mortality-africa-safer-future/>
- World Bank, 2024. *Poverty and Inequality Platform (PIP)*. [Online]
Available at: https://data360.worldbank.org/en/dataset/WB_PIP
[Accessed 5 June 2025].
- World Bank, 2025. *Poverty and Inequality*. [Online]
Available at: <https://datatopics.worldbank.org/world-development-indicators/themes/poverty-and-inequality.html>
- WorldBank, 2021. *Sub-Saharan Africa: Tertiary Education*. [Online]
Available at: <https://thedocs.worldbank.org/en/doc/908af3404023a2c31ef34853bba4fe60-0200022022/original/One-Africa-TE-and-COVID-19-11102021.pdf>

